MATH 1342 – Elementary Statistics – Week 1: Introduction and the Structure of Data

Aug 25 – 27 | Introduction: What Is Statistics For? • 1.1 The Structure of Data | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe statistics as the process of collecting, describing, and analyzing data, and judge a striking result by how likely it is to occur by chance alone. (CLO 1, CLO 4) **2.** Distinguish a sample from a population and explain how a non-representative sample biases an estimate. (CLO 1) **6.** Compute and compare the mean and median of a small dataset and explain how an outlier affects each. (CLO 2)	Introduction Slides: What Is Statistics For? (Day 1)	Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age		Blackboard Zoom (live class)
3. Identify the cases and variables in a dataset presented as a table. (CLO 2) **4.** Classify a variable as categorical or quantitative. (CLO 2) **5.** Identify the explanatory and response variables in a research question. (CLO 1)	e-Text: 1.1 The Structure of Data Interactive Lecture: 1.1 The Structure of Data Slides: 1.1 The Structure of Data	Participation: Variables – identify cases, variables and variable types	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services	Participation: Syllabus – course tour and expectations		Content Area in Blackboard Zoom (live class)

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 2** Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 4** Explain the role of probability in statistics.

MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Sep 1 – 3 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Distinguish a population from a sample and describe statistical inference. (CLO 1) **2.** Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CLO 1) **3.** Recognize bias from question wording, question context, and inaccurate responses. (CLO 1)	e-Text: 1.2 Sampling from a Population Interactive Lecture: 1.2 Sampling from a Population Slides: 1.2 Sampling from a Population (includes sampling video)	Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1 Participation: Sampling – spot the bias in sample scenarios Participation: Sampling Methods (Kahoot!) – Thu: sampling review	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard Kahoot! Zoom (live class)
4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CLO 1) **5.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives)	Assignment Tool in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions.

MATH 1342 – Elementary Statistics – Week 3: Describing Categorical and Quantitative Variables

Sep 8 – 10 | 2.1 Categorical Variables • 2.2 One Quantitative Variable: Shape and Center | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct frequency and relative frequency tables, compute a sample proportion $\hat{p}$, and distinguish the parameter p from the statistic $\hat{p}$. (CLO 2) **2.** Compute and interpret overall and conditional proportions, and a difference in proportions, from a two-way table, choosing the correct denominator. (CLO 2) **3.** Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)	e-Text: 2.1 Categorical Variables Interactive Lecture: 2.1 Categorical Variables Slides: 2.1 Categorical Variables	Participation: Frequency and Relative Frequency Participation: Finding Proportions from Two-Way Tables	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2) **5.** Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CLO 2) **6.** Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median Estimation activity: estimate the mean and SD of a dotplot without calculating (95% rule)	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data.

MATH 1342 – Elementary Statistics – Week 4: Percentiles, Boxplots and Two-Variable Relationships

Sep 15 – 17 | 2.3 One Quantitative Variable: Percentiles • 2.4 Two Variable Relationships | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **2.** Compute and interpret z-scores to compare values measured on different scales or in different groups. (CLO 2) **3.** Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2 Participation: Using Percentiles – Your Turn: comparing heights with z-scores Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	WileyPLUS StatKey Blackboard Kahoot! Zoom (live class)
4. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CLO 2) **5.** Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CLO 7) **6.** Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Quick Self-Quizzes: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM)	WileyPLUS Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.4 (all objectives)	Assignment Tool in Blackboard

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 7** Solve linear regression and correlation problems.

MATH 1342 – Elementary Statistics – Week 5: Linear Regression

Sep 22 – 24 | 2.5 Two Quantitative Variables: Linear Regression • Catch-Up Day (Thu Sep 24) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CLO 7) **2.** Use a regression equation to predict a response value. (CLO 7) **3.** Compute and interpret a residual, and locate positive and negative residuals on a scatterplot. (CLO 7) **4.** Use technology to find the least-squares regression line and explain what "least squares" means. (CLO 7) **5.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Participation: Two Variable Relationships – Tue: correlation review, Your Turn: spotting residuals Participation: Regression Line – Thu: baseball game length and used-car price problems Example 2 (commute distance and time): find the regression line with technology and predict	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 7 Solve linear regression and correlation problems.

MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 29 – Oct 1 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space and count outcomes to compute probabilities for equally likely outcomes, and apply the complement rule. (CLO 3) **2.** Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ, σ, p, ρ vs. $\bar{x}$, s, $\hat{p}$, r). (CLO 5) **4.** Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CLO 5) **5.** Explain the standard error as the variability of a statistic and predict how sample size affects it. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	Online Homework: 3.1 Sampling Distributions (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Take-Home Quiz 3 (due Tue Sep 29, 11:59 PM) – covers 2.5 Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	Assignment Tool in Blackboard Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables.

MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 6 – 8 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval for a mean or proportion as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **2.** Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6) **4.** Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6) **5.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.

MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests

Oct 13 – 15 | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CLO 3, CLO 6) **2.** Predict how changing the confidence level or the sample size affects the width of an interval. (CLO 6) **3.** Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Blackboard
4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CLO 8) **5.** Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming H_0 and asking whether the data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Writing Hypotheses – Rock paper scissors practice (9 of 11 wins): define the parameter, state H_0 and H_a, choose the tail Think about it: is 15 of 16 correct (dolphin study) convincing?	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers P.1, 3.1–3.3	Assignment Tool in Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 20 – 22 | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CLO 5, CLO 8) **2.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **3.** Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a. (CLO 3, CLO 8) **4.** Interpret a p-value in context and relate its size to the strength of evidence against H_0. (CLO 4, CLO 8) **5.** Compare the p-value to α, decide whether to reject H_0, write a two-step conclusion in context, and avoid saying "accept H_0". (CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video) StatKey – Randomization Hypothesis Tests	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed Participation: P-values and Conclusions – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3	WileyPLUS StatKey Blackboard Zoom (live class) Assignment Tool in Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing and Making Connections

Oct 27 – 29 | 4.4 A Closer Look at Testing • 4.5 Making Connections | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8) **2.** Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CLO 8) **4.** Compare bootstrap and randomization distributions by their center and use, identifying each center for given data. (CLO 5) **5.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4, 4.5 (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Practice Test 2 (Blackboard, by Sun Nov 1) Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 11: Normal Distributions and Inference for a Proportion

Nov 3 – 5 | 5.1 Hypothesis Testing Using Normal Distributions • 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution N(null value, SE). (CLO 5) **2.** Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8) **3.** Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **4.** Construct and interpret a confidence interval for a population proportion with z* for 90%, 95%, or 99% confidence. (CLO 6) **5.** Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test) Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM) Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 12: Inference for Two Proportions and for a Mean

Nov 10 – 12 | 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) • 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CLO 5) **2.** Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **3.** Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT) Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. Explain why inference for a mean uses the t distribution with df = n − 1, and check the conditions for t procedures. (CLO 5) **5.** Construct and interpret a t confidence interval for a population mean, and carry out a one-sample t test. (CLO 6, CLO 8)	e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT) Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 5.1, 6.1–6.3 (all objectives)	Assignment Tool in Blackboard

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data

Nov 17 – 19 | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CLO 5) **2.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$, explaining how the order of the groups affects the sign. (CLO 6) **3.** Carry out a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT) Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8) **5.** Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CLO 6, CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13. (all objectives)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association

Nov 24 – 26 | 7.2 Testing for an Association Between Two Categorical Variables • Thanksgiving Holiday – No Class (Thu Nov 26) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **2.** Compute expected counts and check that all expected counts are at least 5. (CLO 8) **3.** Compute the chi-square statistic and df = (r − 1)(c − 1), and interpret the right-tail p-value. (CLO 8) **4.** Relate the chi-square test on a 2×2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.4, 6.5 Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)	Assignment Tool in Blackboard Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 8 Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation

Dec 1 – 3 | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CLO 8) **2.** Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0. (CLO 8) **3.** Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CLO 8)	e-Text: 8.1 Analysis of Variance Interactive Lecture: 8.1 Analysis of Variance Slides: 8.1 Analysis of Variance StatKey – Theoretical Distributions (F)	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. State the simple linear model, distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r, and interpret the slope in context. (CLO 7) **5.** Use regression output to test H_0: $\beta_1 = 0$ (equivalently $\rho = 0$) with a t statistic on $n - 2$ degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CLO 7, CLO 8) **6.** Construct and interpret a confidence interval for the slope, interpret R^2 as the proportion of variability explained, and check the conditions with a residual plot. (CLO 6, CLO 7)	e-Text: 9.1 Inference for Slope and Correlation Interactive Lecture: 9.1 Inference for Slope and Correlation Slides: 9.1 Inference for Slope and Correlation	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); self-quiz: drinks per week vs. classes missed	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)

CLO 6 Describe and compute confidence intervals. **CLO 7** Solve linear regression and correlation problems. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 16: Putting It All Together and Final Exam

Dec 8 – 10 | Review: Putting It All Together – From Sampling to Hypothesis Testing (Tue Dec 8) • Final Exam – Thu Dec 10, online (Blackboard, proctored) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Choose the correct procedure among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression, and carry out a complete six-step hypothesis test. (CLO 8) **2.** Carry out and interpret a complete confidence interval or hypothesis test, using simulation or theoretical distributions as appropriate. (CLO 6, CLO 8) **3.** Interpret results in context, including what the study design allows the conclusion to claim about generalization and causation. (CLO 1)	Review Slides: Putting It All Together (with answer key and no-answers versions) All Section Slides and Interactive Lectures (Weeks 1–15) StatKey	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	Final Exam – Thu Dec 10 (Blackboard, proctored, 2 hours, cumulative; no retake)	Blackboard StatKey Zoom (live class) Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.